

BRAC UNIVERSITY
Department of Computer Science and Engineering

Set: B

Examination: Final
 Duration: 2 Hours

Semester: Summer 2025
 Full Marks: 60

CSE 340: Computer Architecture

Answer the following questions. Show calculations where required. Figures in the right margin indicate marks. Understanding the question is part of the examination.

	ID:	Section:
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1. 2+
2+
2-

CO1 a) In three different arbitrary-length IEEE 754 floating-point representations, consider the exponent fields to be 8, 10, and 12 bits long, respectively. If the biased exponents are 25, 80 and 120, respectively, then **calculate** the actual exponents of the given biased exponents. Your answer should show the detailed calculations. 2-

b) **Calculate** the smallest positive non-zero decimal number in IEEE 754 single precision representation? 3

c) **Why** does IEEE 754 use a hidden leading 1 for normalized numbers, but not for denormalized numbers? 2

d) **Write** the name and syntax of the floating-point instructions for the following: 1+
1

 - i. Converting from integer to float.
 - ii. Converting from float to integer..

e) **Construct** the equivalent RISC-V assembly code for the following: 2.5
 Suppose you have two double precision numbers stored in f_1 and f_2 , floating point registers. If the value of f_1 is less than f_2 , then store 0 in X_1 else store 1.

f) Consider $X = 3.214$ and $Y = 4.876$, **perform** $X*Y$ using IEEE754 single precision floating point representation. **Mention** if there is any overflow or underflow. Your answer should show all the detailed steps and the final decimal value. Consider 5 digits after the floating point while performing the conversions. 2+
0.5
2. 2+
2

CO2 a) In case of a Branch instruction, **why** does RISC-V split immediate fields across different parts of an instruction? **Encode** the instruction BEQ $X_5, X_6, 48$; Use don't care (X) for fields such as opcode and funct3. 2

b) **What** is the main purpose of the LUI instruction, and how is it typically used in relation to immediate values? 2

c) Consider the below instruction set: 1+
2

PC	Instruction
	LUI $X_5, 0x30000$
0x1000	JALR $X_1, 0(X_5)$

 - i. As we are using X_1 register instead of X_0 with the JALR instruction, **what** is the return address stored in X_1 for JALR instruction?
 - ii. **Find** the value of the target address to jump from JALR instruction.

d) Why can't shifting right be used instead of division in RISC-V all the time?

2

3. CO3 a) Consider the following instruction mix given below. In a single cycle Datapath scenario, answer the following questions:

2+
2+
2

R-type	I-type (non-lb)	Load	Store	Branch	Jump
10%	25%	15%	15%	20%	15%

- For **what** fraction of all instruction register write signals are enabled?
- For **what** fraction of instructions does the final output come directly from the ALU?
- For **what** fraction of all instruction data needs to perform sign extension?

b) Design a single-cycle Datapath for the following instruction. Your diagram should have labeling of bit lengths and data flow along with the control signals. Please note that you have to design a Datapath that can only execute the instructions below. Designing a general Datapath will have a severe penalty.

5

BNE X₃, X₄, Else

c) Explain the following statements:

2+
2+
2

- BEQ can generate control hazard.
- A data hazard can occur between two logical instructions.
- LD data hazard needs a stall and then forwarding to overcome it.

d) In the table below, we examine how pipelining affects the clock cycle time of the processor. This problem assumes that individual stages of the datapath have the following latencies:

2+
2

IF	ID	EX	MEM	WB
260ps	220ps	260ps	200ps	220ps

- What is the clock cycle time in a pipelined and non-pipelined processor?
- What is the total time required of an instruction set having 5 instructions? Consider there are two data hazards, and each needs 2 stalls to overcome the hazards.

e) SUB X₂₂, X₂₇, X₂₇
LD X₂₂, 16(X₂₇)
SD X₂₂, 16(X₂₇)

2+
2

For the above instruction set running in a pipelined processor,

- Solve the data hazard using only Forwarding.
- Find the total number of clock cycles required to run the instruction set.

4. CO4 a) Find the AMAT for a processor with a 3 ns clock cycle time, a miss penalty of 25 clock cycles, a miss rate of 0.06 misses per instruction, and a cache access time (including hit detection) of 1.1 clock cycles.

2

b) LUI X5, 0x10010 # Base of array A
 LUI X6, 0x20020 # Base of array B
 ADDI X7, X0, 100 # Loop bound = 100 elements
 ADDI X8, X0, 0 # i = 0
 ADDI X10, X0, 0 # sum = 0

Loop:

LD X11, 0(X5) # Load A[i]
 LD X12, 0(X6) # Load B[i]
 ADD X13, X11, X12 # temp = A[i] + B[i]
 ADD X10, X10, X13 # sum += temp
 SD X13, 0(X5) # Store result back to A[i]
 ADDI X5, X5, 8 # Move to next A[i+1]
 ADDI X6, X6, 8 # Move to next B[i+1]
 ADDI X8, X8, 1 # i++
 BLT X8, X7, Loop # Repeat

Consider the instruction sequence above and answer the following questions:

- Why can the addi instruction ADDI X₈, X₈, 1 be called temporal locality?
- How does the load instruction LD X₁₁, 0(X₅) benefit from temporal locality?